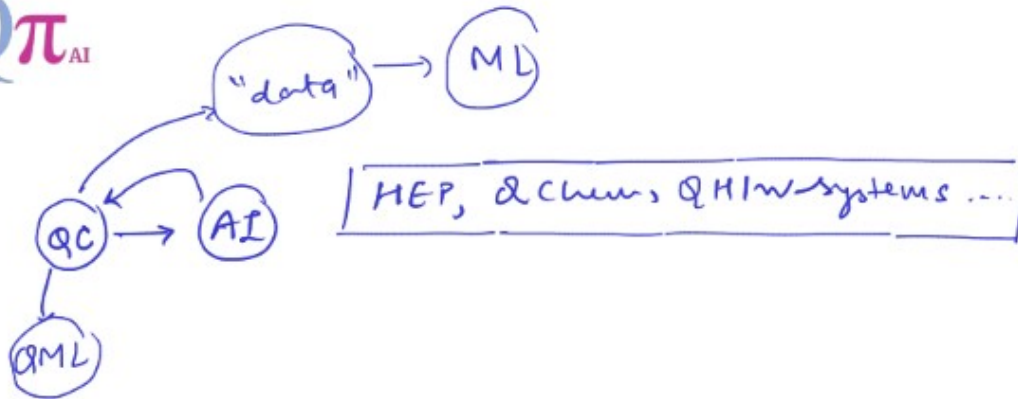
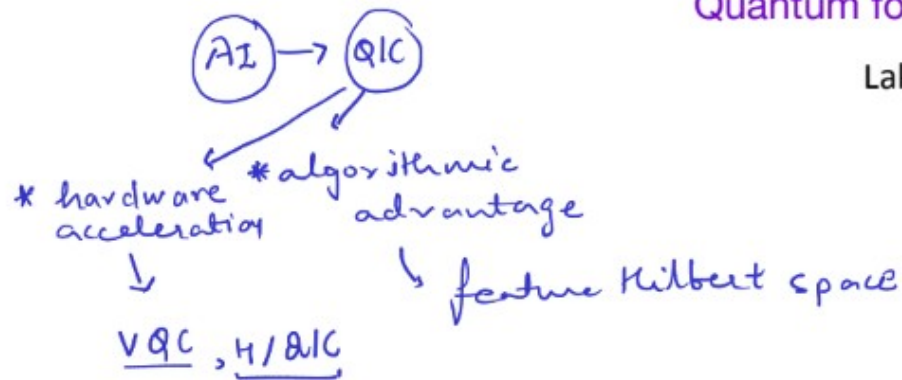


Quantum Machine Learning

Quantum for AI and AI for Quantum

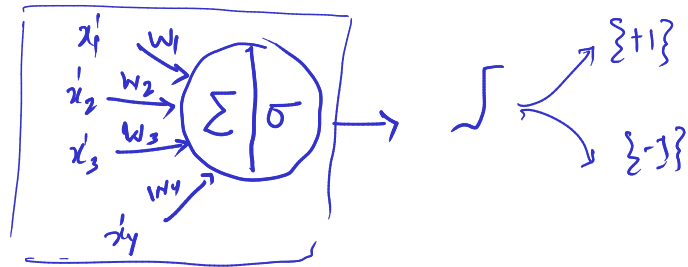
Lakshya Priyadarshi



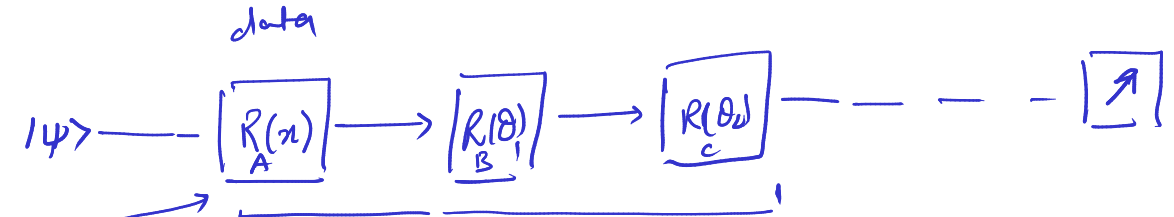
$$\vec{\theta} = \{\theta_1, \theta_2\}$$

classical Neural N/w

$$\hat{y} = \sigma(w^T \vec{x} + b)$$



quantum NN (1 qubit)

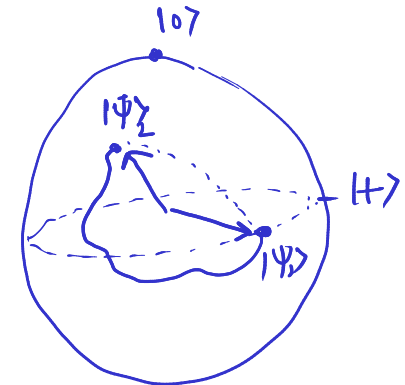


$$\vec{X} = [0.5]$$

"angle-encoding"

→ "amplitude encoding" -----

→ "phase encoding" -----



A single-qubit neural network classification model

- **Target states**

Each class is represented by one of the maximally orthogonal points

- **Parameterized quantum circuit**

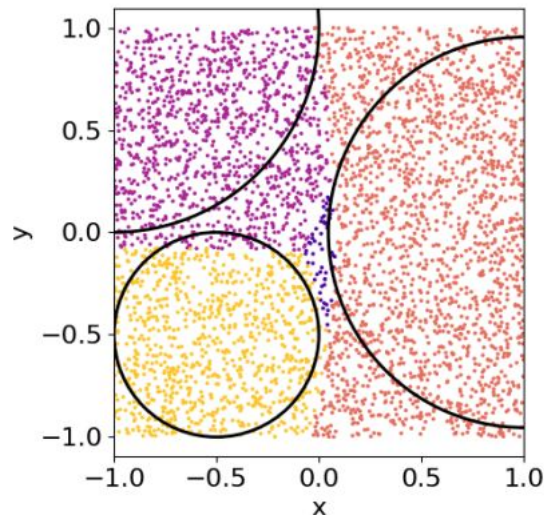
Encode data-point and encode trainable parameters in circuit

- **Loss function**

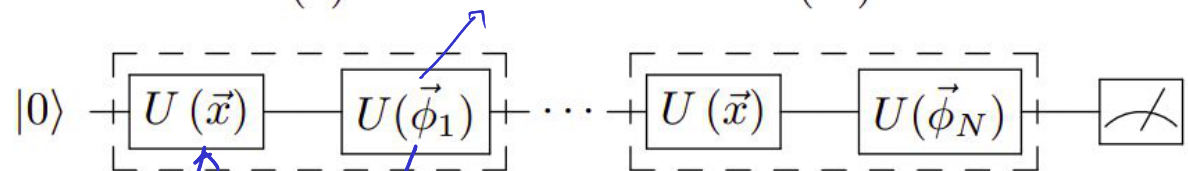
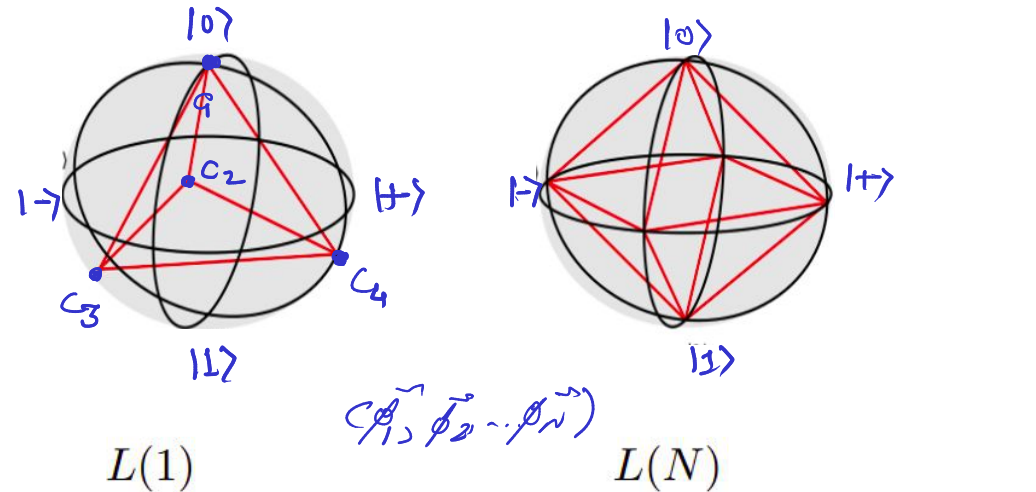
Overlap (target state, output state) for all training points

- **Quantum classifier**

Classify data points based on qubit states



$$|\psi^*\rangle \rightarrow \{c_1, c_2, c_3, c_4\}$$



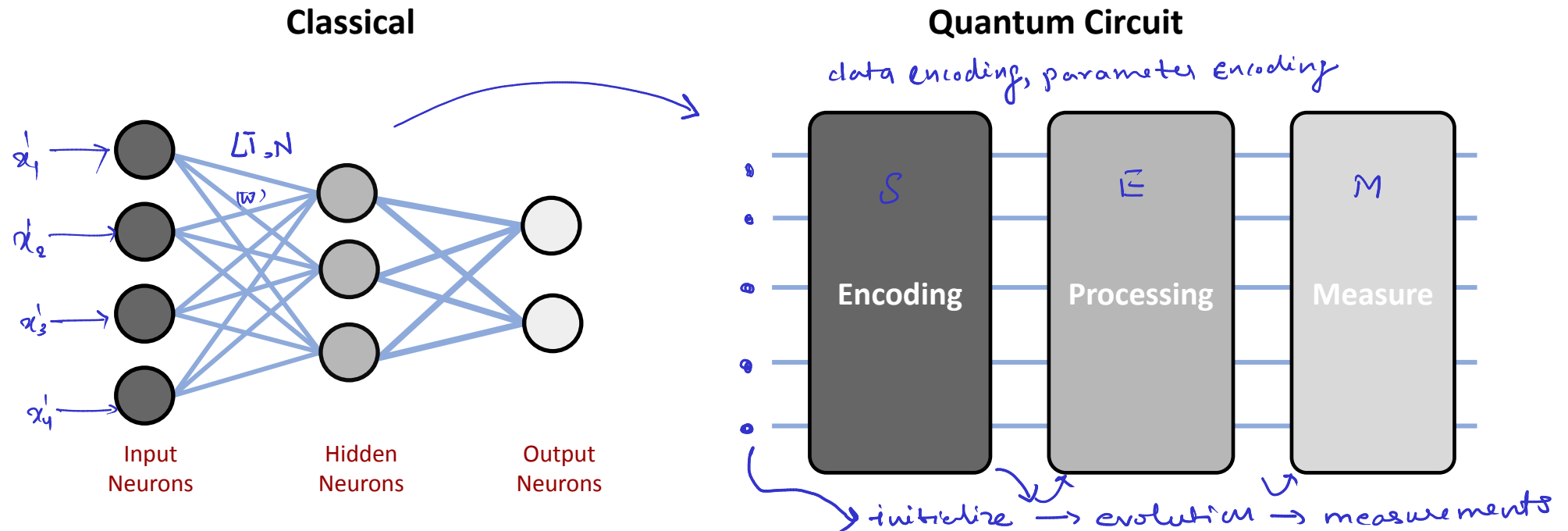
$$\chi_f^2(\vec{\theta}, \vec{w}) = \sum_{\mu=1}^M \left(1 - |\langle \tilde{\psi}_s | \psi(\vec{\theta}, \vec{w}, \vec{x}_\mu) \rangle|^2 \right)$$

actual state

REF: Data re-uploading for a universal quantum classifier, <https://arxiv.org/pdf/1907.02085.pdf>

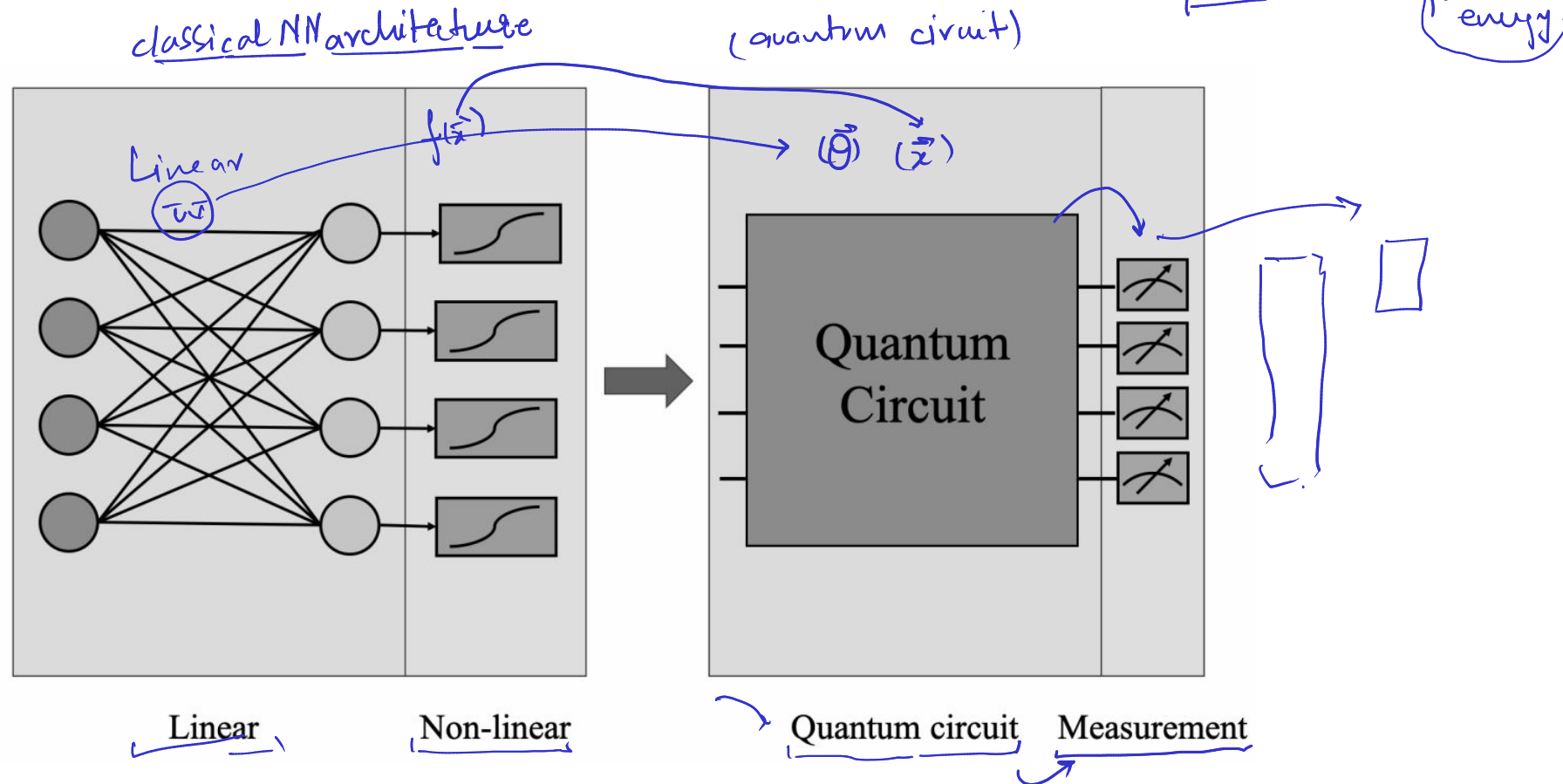
Building neural networks in the quantum model

$$f(\vec{\theta}, \vec{x}) \rightarrow f_a$$



REF: M. Schuld, A. Bocharov, K. M. Svore, and N. Wiebe, Phys. Rev. A 101, 032308 (2020)

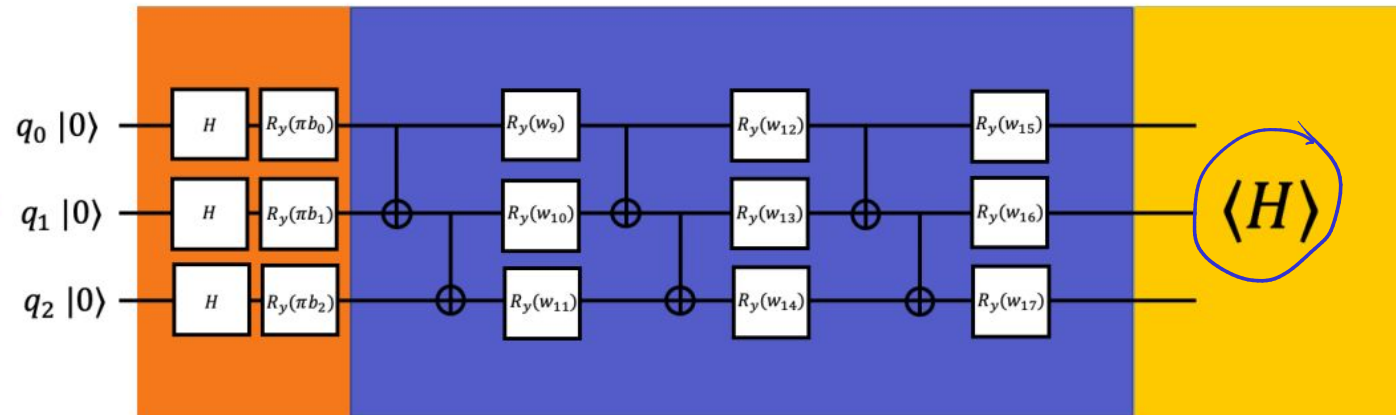
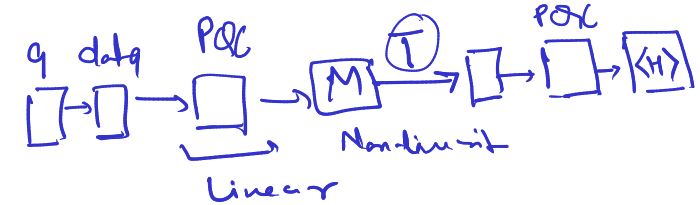
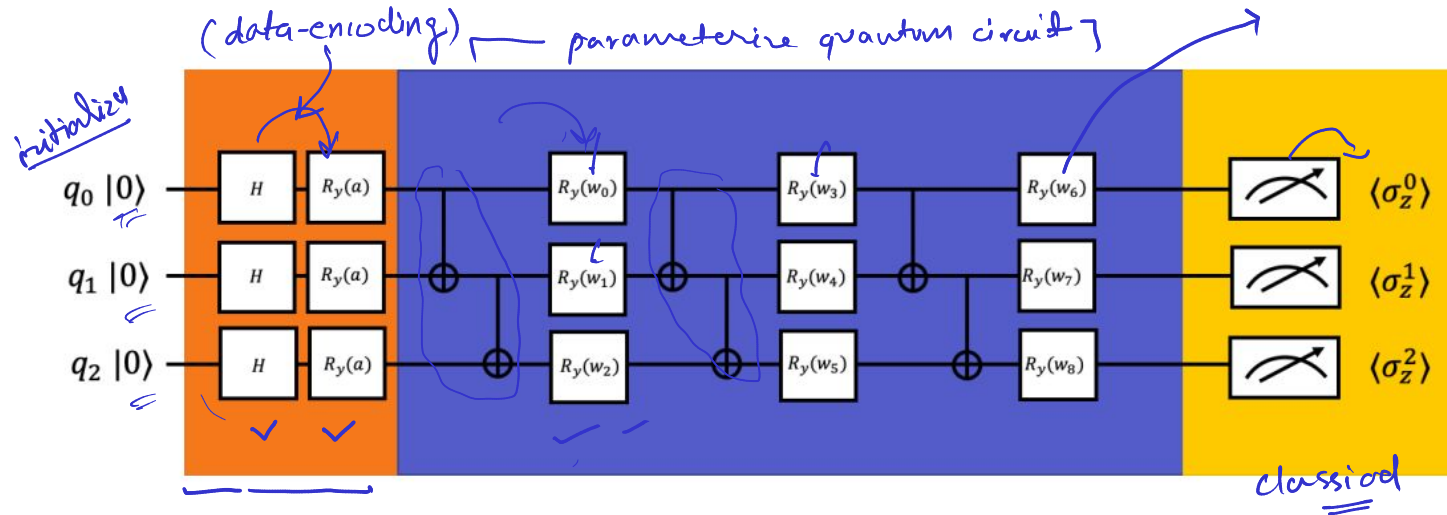
Hybrid quantum-classical neural networks



✓ REF: Hybrid Quantum-Classical Neural Network for Calculating Ground State Energies of Molecules,
<https://arxiv.org/pdf/1912.06184.pdf>

Hybrid quantum-classical neural networks

$a \rightarrow$ input (bond length)



REF: Hybrid Quantum-Classical Neural Network for Calculating Ground State Energies of Molecules,
<https://arxiv.org/pdf/1912.06184.pdf>

Hybrid quantum-classical neural networks

$$\{a\} \xrightarrow{\text{Train}} \rightarrow$$

• Input $|0\rangle^{\otimes n}$

- Data encoding: $\{\underline{a_0}, \underline{a_1}, \dots, a_{n-1}\}$, $G = \bigotimes_{i=0}^{n-1} g_i(f_i(a_i))$

$$|\psi\rangle = \bigotimes_{i=0}^{n-1} (R_y(a_i) H) |0\rangle^{\otimes n}$$

- Parameter encoding: $\{\theta_0, \theta_1, \dots, \theta_{k-1}\}$

$$R_x(\theta) = e^{-i\theta/2 \sigma_x}$$

$$U(\vec{\theta})|\psi\rangle = \left(\prod_i V_i\right)|\psi\rangle$$

- Quantum layer:

PQC

$$\bigotimes_{\text{all qubit}} (R_k \text{ CNOT}_{\text{blw}(k, l)})$$

- Classical layer:

$$\langle \sigma_z^i \rangle \rightarrow \text{Next: } \bigotimes_{i=0}^{n-1} R_y(b_i \pi) H \quad [-1, 1] \rightarrow [-\pi, \pi]$$

- Cost Function:

$$f = \sum_j \langle \phi_j | H_j | \phi_j \rangle$$

similar VQE $\rightarrow$ classical optimizer

Quantum NN $\rightarrow$ unsupervised learning setting



Classical Problems	Quantum Tools	Applications	Methods
BLAS → matrix inversion, inner products, eigenvalue decomposition, singular value decomposition	QBLAS/HHL → quantum phase estimation, post selective amplitude update, Hamiltonian simulation, density matrix exponentiation	Support vector machines, Gaussian processes, linear regression, discriminant analysis, recommendation systems, principal component analysis	QSVM , QPCA, LSE Regression, Quantum Gaussian Processes
Search → finding closest neighbors, Markov chains	Grover Search → amplitude amplification, quantum walks	k-nearest neighbor, page ranking, clustering, associative memory, perceptrons, active learning agents, natural language processing	Quantum K-means clustering → <i>k-medoid</i>
Sampling → sampling from the model distribution	Quantum Sampling → quantum annealing, quantum rejection sampling	Boltzmann machines, Bayesian nets, Bayesian inference	Bayesian Inference, Quantum Boltzmann Machines
Combinatorial Optimization → combinatorial optimization, <u>QUBO</u> problems	Ground State Optimization → adiabatic quantum computing, quantum annealing, quantum simulation	associative memory, boosting, debugging, variational Bayes inference, Bayesian networks, perceptron, EM algorithm, clustering	Bayesian Inference, Perceptron

* use qc to optimize certain subprocedures in M
* construct qml models by converting
ML → quantum form quantum SVM ⇒ $K(i,j)$
 $\langle \psi | \phi \rangle \rightarrow$

machine learning method

quantum approach

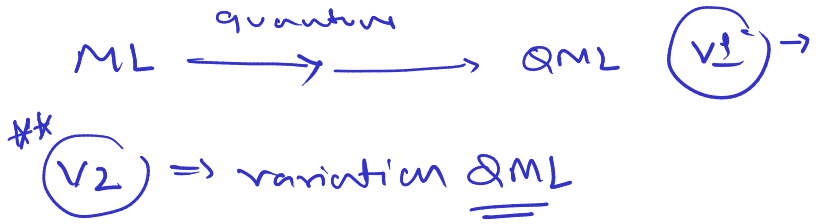
- k-nearest neighbour
- support vector machines
- k-means clustering
- neural networks
- decision trees
- Bayesian theory
- hidden Markov models

Efficient calculation of
classical distances on a
quantum computer ✓

First explorations of
quantum models

Reformulation in the
language of open
quantum systems

REF: An introduction to quantum machine learning, M. Schuld, 2014



* Q SVM
* Q KNN
* Q clustering

data
⇒ procedures.

L-S

Method	Speedup	AA	HHL	Adiabatic	QRAM
Bayesian Inference [107, 108]	$O(\sqrt{N})$	Y	Y	N	N
Online Perceptron [109]	$O(\sqrt{N})$	Y	N	N	optional
Least squares fitting [9]	$O(\log N^{(*)})$	Y	Y	N	Y
Classical BM [20]	$O(\sqrt{N})$	Y/N	optional/N	N/Y	optional
Quantum BM [22, 62]	→ $O(\log N^{(*)})$	optional/N	N	N/Y	N
Quantum PCA [11]	→ $O(\log N^{(*)})$	N	Y	N	optional
→ Quantum SVM [13]	$O(\log N^{(*)})$	N	Y	N	Y
Quantum reinforcement learning [30]	$O(\sqrt{N})$	Y	N	N	N

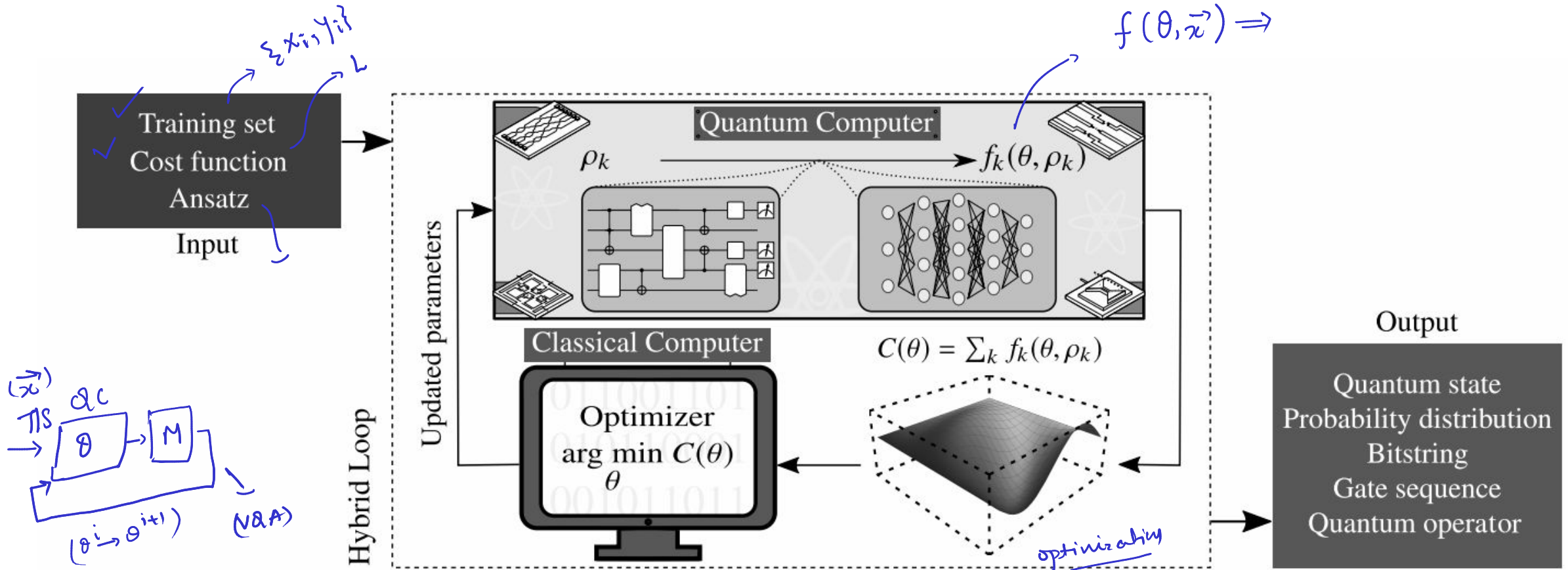
Quantum Memory

↓
Efficiently
prepare
quantum
states.

"theoretical"

REF: Quantum Machine Learning, Biamonte, et. al. <https://arxiv.org/pdf/1611.09347.pdf>

$$f(\theta, \vec{x}) \Rightarrow$$



REF: Variational quantum algorithms, M. Cerezo et. al.

What are we learning in this lecture?

→ Quantum Models of classical ML
✓ → QuantumClassical → NISQ

* why AI for Quantum, Quantum for AI?

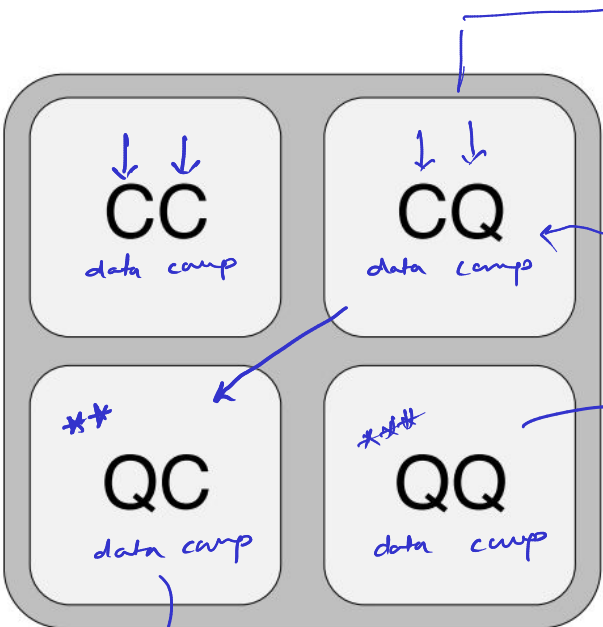
classical data
"0,1,1,0..."

"qubit"
2 level

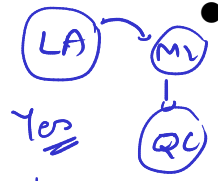
quantum data

→ superpos
→ interfere
→ quantum correlations

Problem/Data



Algorithm/computer



Mathematical framework

- Machine Learning - Linear Algebra
- Quantum Computing - Linear Algebra

Variational Quantum Algorithms

- Parameterized circuits $(\gamma, \nu), \theta$
- Optimization of parameters

data features
High-dimension V.S.

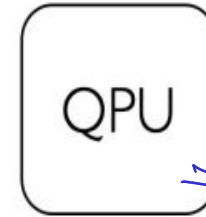
$$|\psi\rangle = \begin{bmatrix} |0\rangle \\ |1\rangle \\ |0\rangle \\ |1\rangle \end{bmatrix} \rightarrow \begin{bmatrix} x \\ p \end{bmatrix}$$

$H|\psi\rangle$

Particle Traj. Reconstruction using Quantum Graph NMs



γ, ν



state prep

$$\langle H_C \rangle_{\gamma, \nu}$$

QML
↓
Quantum Models.
(advantages)

* "Optimize quantum control using reinforcement learning"

Physics Experiments (HEP) → Particle colliders

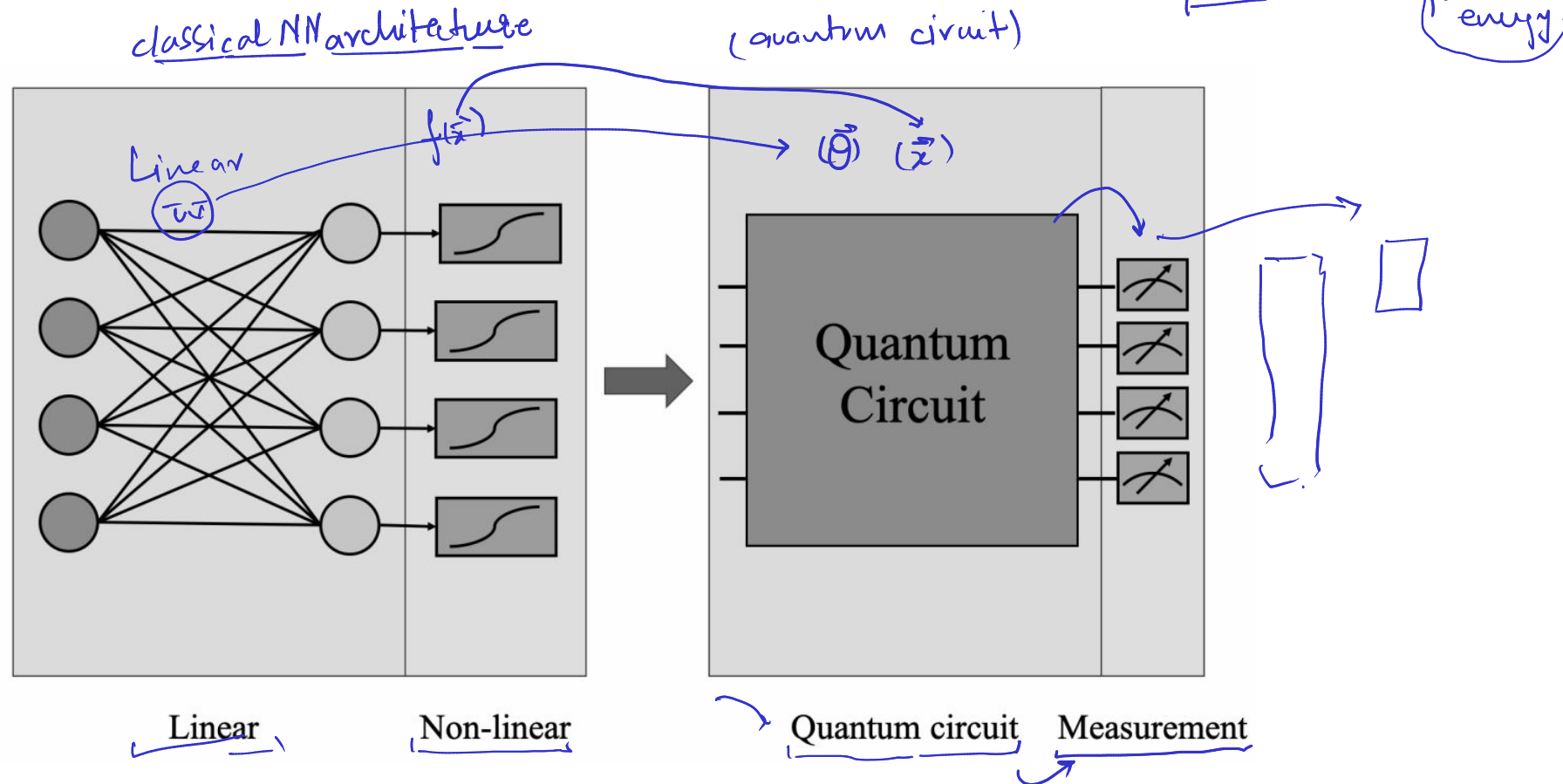
Quantum → Molecular structure

Drug → ...

QC

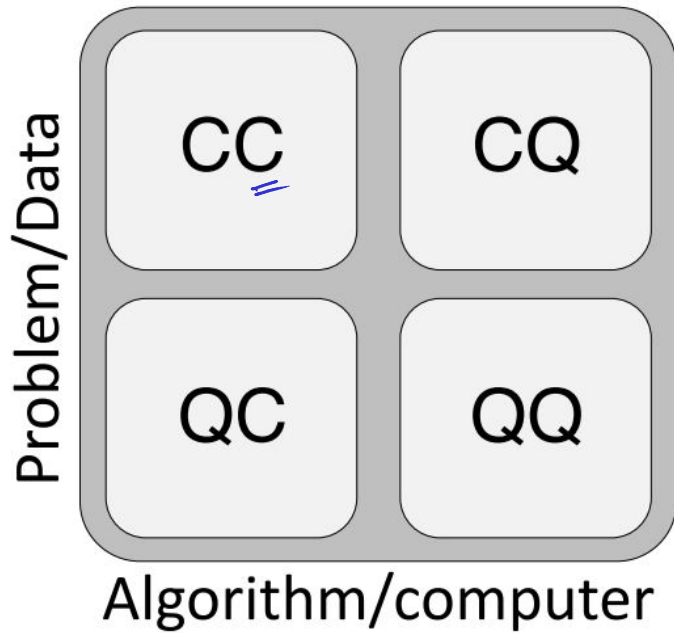
- * Data Analysis of HEP systems
- * DFT-assisted ML

Hybrid quantum-classical neural networks



✓ REF: Hybrid Quantum-Classical Neural Network for Calculating Ground State Energies of Molecules,
<https://arxiv.org/pdf/1912.06184.pdf>

Quantum M/C Learning ① AI/ML ↔ Quantum



→ LA-based QML
 → variational quantum → hybrid Q/C
 → PQC, single-qubit NN
 → nQ/C QNN
 → Quantum-inspired algorithms
 (Recommender System) Tang et al.
 → QCNN
 → MNIST

